Sensor-Free Ego-Motion Compensation via Full-Affine ORB for Egocentric Pedestrian Prediction

Anonymous Authors
Paper under blind review — remove for camera-ready

***Abstract*—Egocentric pedestrian tracks mix pedestrian motion with camera motion. Zhang et al. (TIP 2024) decompose using ego speed as input and soft supervisor; LIM (TITS 2025) deletes speed to avoid causal confusion. We study *sensor-free* compensation with offline ORB partial affines. Learned multi-agent residuals with OBD-magnitude or ORB-regression losses fail a standing-pedestrian variance gate and hurt ADE on PIE. Fixed full-affine ORB compensation passes the gate (∼93% flatten on standing pedestrians while the vehicle moves) and is ADE-neutral vs. a bbox-only GRU (60.5 vs. 60.8 px). GT speed helps intent but hurts ADE, supporting LIM. On JAAD (no OBD) the same protocol yields ADE 62.7 vs. 83.1 px with a 96.9% standing+moving flatten rate.**

I. Introduction

Moving-camera contamination of image-plane coordinates is first-order for ego-view trajectory and intent prediction. Sensor-based fixes are unavailable on JAAD and can leak driver reaction on PIE. We ask whether classical ORB geometry can recover camera motion without a speed sensor at inference.

Contributions: (1) Full partial-affine ORB ego-motion at the pedestrian centre, not global translation. (2) A standing-pedestrian variance gate exposing failure modes of magnitude–speed supervision. (3) Fixed ORB ADE-neutral on PIE and ADE-helpful on JAAD (no OBD). (4) Evidence that GT speed helps intent / hurts ADE (LIM-aligned).

II. Related Work

Zhang et al. use two towers with speed-weighted action-aware loss (ADE 17.41 on PIE with I3D). **LIM** removes speed as a causal confounder. **Ours** recovers geometry offline; ORB is never consumed as a live speed feature.

III. Method

Fixed ORB-affine. Offline ORB + RANSAC `estimateAffinePartial2D` on consecutive frames with pedestrian boxes masked. For ego centre (c_x, c_y), transition target is $M[c_x, c_y, 1]^\top - [c_x, c_y]^\top$. Cumsum compensates observation and future; GRU predicts in stabilized frame; metrics in ego view.

Variance gate. For standing pedestrians while the vehicle moves, valid compensation requires $\mathrm{Var}(\mathrm{comp}) < \mathrm{Var}(\mathrm{raw})$ on observed centres.

Learned LEMC (ablation). Multi-agent displacement statistics → GRU → residual → cumsum subtract. Fails gate under joint training.

TABLE I
PIE test (mean±std, 3 seeds).

Method	ADE	FDE	AUC	F1
Baseline	60.8±3.3	121.8±9.9	0.89±0.01	0.74±0.03
GT speed	67.0±1.3	136.9±3.0	0.93±0.01	0.85±0.02
LEMC+OBD	85.8±11.2	164.3±11.3	0.80±0.01	0.60±0.03
Fixed ORB	**60.5**±**1.8**	130.3±1.3	0.80±0.01	0.60±0.01

IV. Experimental Setup

PIE (augmented track store, 1773 test windows) and JAAD (323 videos, ∼295 test windows with TTE). Metrics: ADE/FDE, ARB/FRB@30f, AUC/F1. Three seeds; mean±std reported. Bbox-only GRU backbone — not comparable to Zhang's I3D ADE.

V. Results

Table I summarizes PIE. Fixed ORB matches baseline ADE; learned LEMC variants are worse. GT speed improves AUC (0.93) but degrades ADE (67.0). Sign gate: attempt-3 OBD aux 0.8% vs. fixed ORB 93.1% standing+moving flatten.

JAAD (n=295 test windows, 3 seeds): baseline ADE 83.1±2.0 px / FDE 156.3±4.3 px; fixed ORB ADE 62.7±2.3 px / FDE 120.1±2.8 px. Standing+moving flatten 96.9% (all windows 71.9%). Unlike PIE (ADE-neutral), JAAD benefits from affine compensation—consistent with stronger camera motion in the JAAD clip set.

VI. Limitations

Single GRU backbone; ORB depth confound; hold-last for missing future affines; no on-vehicle latency study. We do not claim SOTA ADE vs. vision-heavy baselines.

VII. Conclusion

Sensor-free ego-view compensation is feasible with full-affine ORB and passes a standing-pedestrian gate, matching bbox-only ADE. Magnitude–speed losses and learned residuals do not substitute for directional geometry.

References

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